TRIPWIRE

AI / NLP RISK ENGINE · S&P GLOBAL × CRISIL CODE TO CONNECT 2026

Engineering the clarity to act.

News, SEC filings, central-bank releases and social posts in; explainable risk signals and stress-tested capital impact out. Every number traceable to its source.

THE PROBLEM

- **Noise:** thousands of items a day; the few that matter (a deposit run, a default, a sanction) look like everything else.
- **No link to the book:** news stops at the headline; nobody connects the event to the loans, bonds and equity it moves.
- **Not defensible:** a score with no evidence, assumptions or measured accuracy cannot go to a credit committee.

Tripwire answers, for every event: what happened, how much it matters, what it does to this bank's capital, and why we believe it.

70.3%
event accuracy on unseen, human-labelled articles

0.72
Impact AUC on 1,003 out-of-time 8-K filings

680
human labels behind every accuracy claim

6
live public sources, collected every 10-30 min

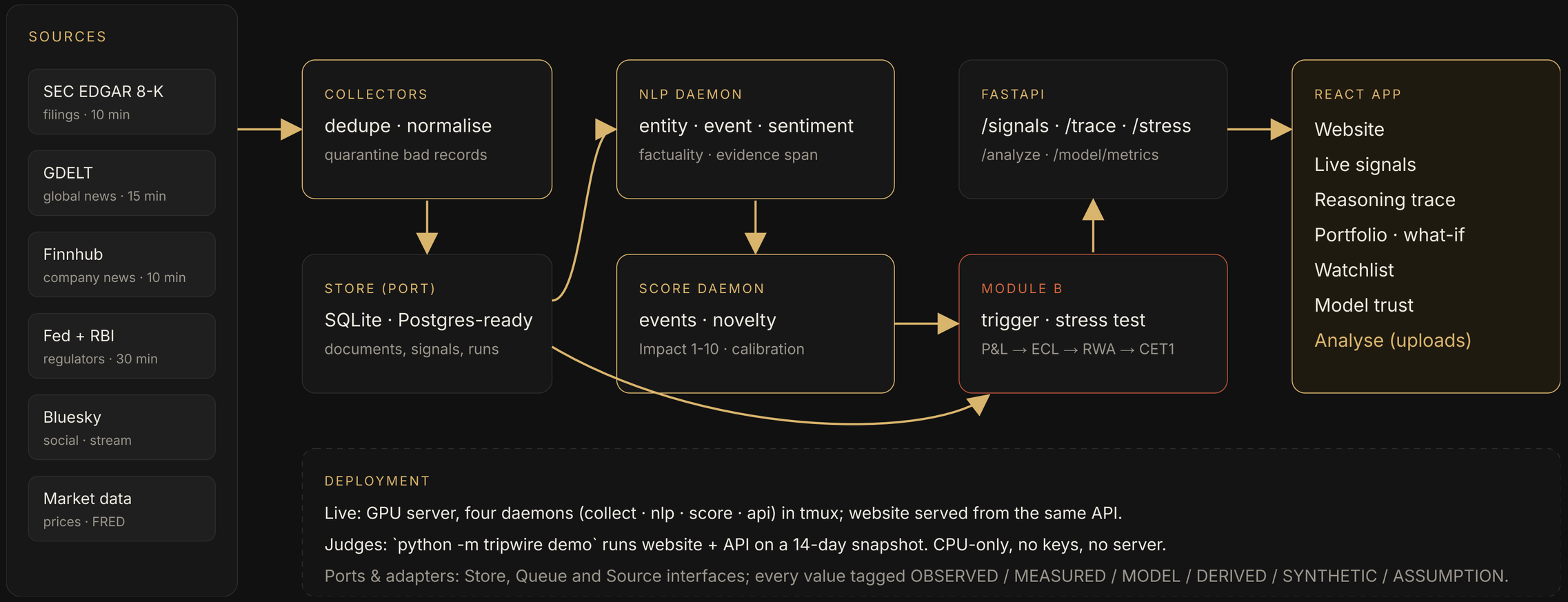
DEEPAK MEENA · IIIT DELHI · DEEPAK23188@IIITD.AC.IN

One engine, signal to capital.

MODULE A · RISK ENGINE: SUBJECT, EVENT + EVIDENCE, SENTIMENT, FACTUALITY, IMPACT 1-10

MODULE B · STRESS TESTING: P&L, ECL, RWA, CET1

WEBSITE · DASHBOARD · ANALYSE YOUR OWN DATA



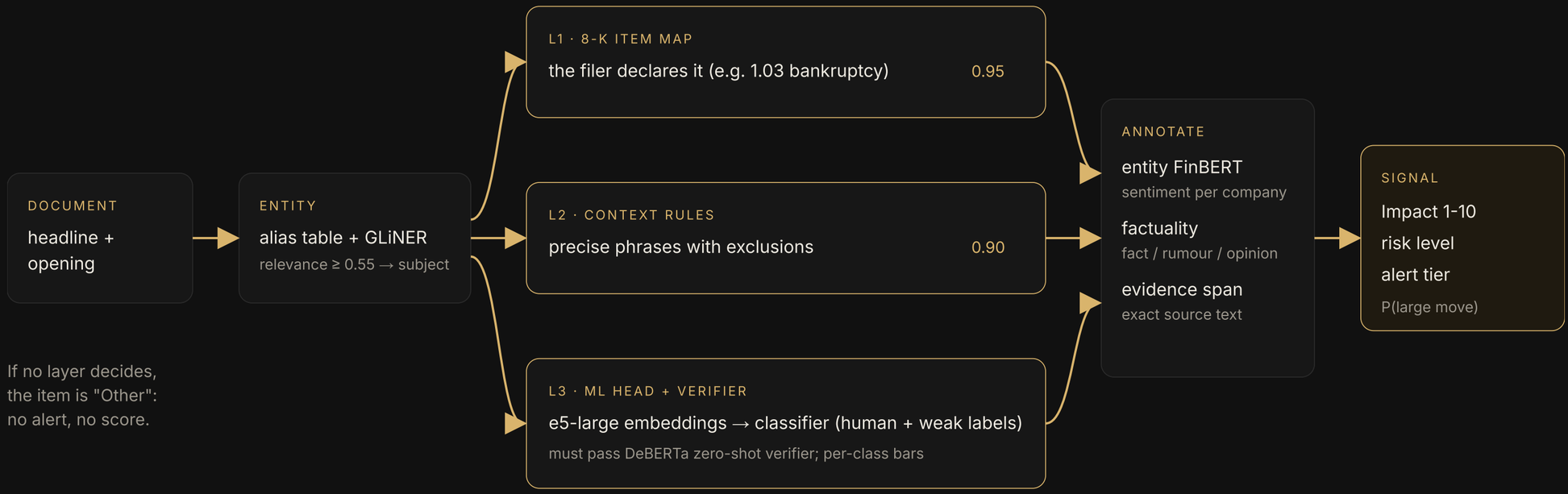
DEPLOYMENT

Live: GPU server, four daemons (collect · nlp · score · api) in tmux; website served from the same API.

Judges: `python -m tripwire demo` runs website + API on a 14-day snapshot. CPU-only, no keys, no server.

Ports & adapters: Store, Queue and Source interfaces; every value tagged OBSERVED / MEASURED / MODEL / DERIVED / SYNTHETIC / ASSUMPTION.

Strongest evidence decides first, the model only fills gaps.



If no layer decides, the item is "Other": no alert, no score.

IMPACT SCORE · SEVEN NAMED FACTORS

SEVERITY 0.25 · SENTIMENT 0.15 · ENTITY SIZE 0.15 · HISTORICAL REACTION 0.15 · NOVELTY 0.10 · CORROBORATION 0.10 · CREDIBILITY 0.10

weighted sum → relevance gate → isotonic calibration to 1-10 and P(large price move); every factor shown on every signal

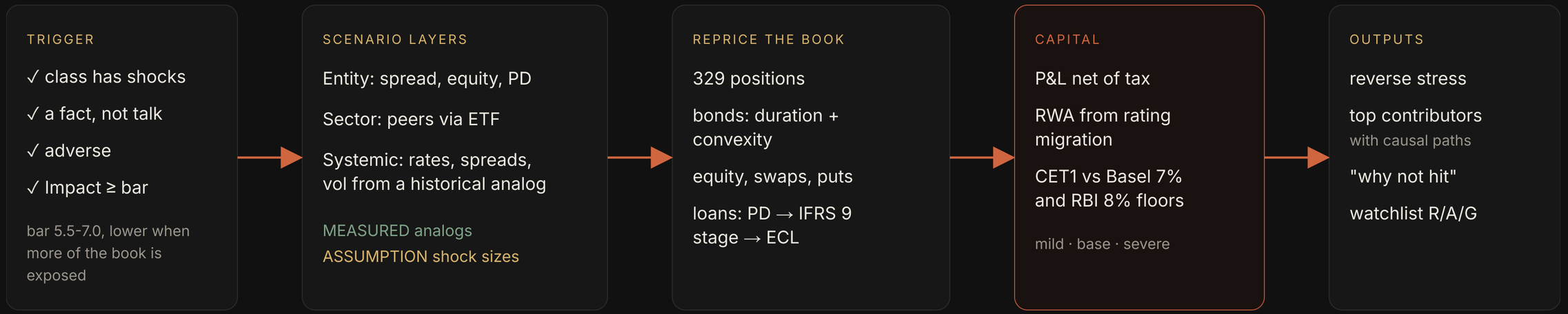
0.72

AUC for large moves on 1,003 out-of-time 8-K filings

0.89

entity-level sentiment macro-F1 (plain FinBERT 0.66)

Stress tests that show exactly what moves CET1.



$451M
synthetic book: 300 SME loans (from Kaggle merchants) + 29 corporate positions

$363M
risk-weighted assets (standardised weights, A-22)

12.0%
starting CET1 ratio ($43.6M, A-21)

28
documented assumptions, each with its rationale

Western Alliance, 13 March 2023: a reassuring statement during a run.

SOURCE

SEC 8-K

Item 7.01, filed 14:58 UTC

EVENT

Credit Event

liquidity stress · rule: bank run · "deposit outflows"

SENTIMENT

+0.86

the text reads positive; the event is adverse by nature

IMPACT

7.4 / 10

96th percentile · P(large move) 44%

STRESS

triggered

Impact clears the exposure bar (6.0)

ASSET CLASS	STRESS P&L
Bonds (rates + bank spreads)	-$1.56M
Equity (regional banks, KRE −23%)	-$0.89M
Corporate loan to WAL (ECL)	-$0.03M
Index put + swaps (hedges)	+$0.33M

RESULT ON THE BOOK

−$2.15M

across 25 positions

CET1 12.00% → 11.59%

above both floors; kept as an end-to-end regression test

Every claim is measured on data the model never saw.

EVENT CLASSIFIER (HUMAN LABELS)	N	ACCURACY	MACRO-F1
Locked test, before → after round 2	212	61.8 → 66.0%	0.59 → 0.64
Fresh out-of-time test 1	182	63.2%	0.61
Fresh out-of-time test 2 (current)	202	**70.3%**	**0.69**

- Company resolution 70.3% · alert precision (standard tier) 74.4%
- Entity-level sentiment: macro-F1 0.89 vs 0.66 for plain FinBERT (SEntFiN test)
- Impact: AUC 0.72, Brier 0.139 on 1,003 out-of-time filings

HOW WE KEEP IT HONEST

- Each fresh test: articles after the model freeze, labelled by a person, evaluated once
- Ship a model only if it beats the current one in cross-validation
- LLM labels only for training and agreement checks (kappa 0.67)

PUBLISHED LIMITS

- "Confirmed" tier: 3/6 on fresh test 2; rules fooled by "launches a dividend", "buyers flood in"
- Weakest on new data: Cyber/Operational recall 0.53, Product Launch precision 0.55
- Impact validated on firm-specific filings only

Try it on your own data, in two commands.

DASHBOARD

- Live signals ranked by risk
- Full reasoning trace per signal
- Portfolio, what-if, watchlist
- Model trust and data sources

ANALYSE

- Article or PDF → full trace + stress run
- Up to 300 headlines → scored table
- Your portfolio vs this week's signals
- Never stored · PDF report to keep

RUN IT

```
pip install -r requirements.txt
python -m tripwire demo
```

CPU-only · no keys · real 14-day snapshot. Open source, MIT.

LINKS

Code `LINK_GITHUB`

Live website `LINK_WEBSITE`

NEXT

More human labels for the rare classes, end-to-end fine-tuning of the encoder, and the confirmed-tier rule fixes, each proven on a new fresh test before it ships.